

Lab: The Whispering Gallery -- How We Synchronize Our Models

Objective

Explore how **communication** works as a process of synchronizing internal models between agents, and investigate why miscommunication happens when models diverge.

Prerequisites

- Completed modules on Systems through Learning
- Read the module on Model Synchronization

Part 1: Telephone Game Analysis

Goal: Experience how generative models drift when signals pass between multiple agents.

Play the telephone game (also called "broken telephone") with at least 4 people:

1. Person 1 whispers a detailed sentence to Person 2.
2. Each person whispers what they heard to the next person.
3. The last person says the sentence aloud.

Record the original and final sentences:

Stage	Message
Original message	
Person 2 heard	
Person 3 heard	
Person 4 heard	
Final message	

Analysis: 1. Where did the biggest distortion occur? 2. Why did each person's generative model "fill in" different details? 3. How does this relate to the Active Inference idea that perception is prediction?

Write your response here

Part 2: Text Message Misunderstanding Lab

Goal: Analyze how communication channels affect model synchronization.

Collect (or create) three examples of text messages that were misunderstood. For each:

Feature	Message 1	Message 2	Message 3
What the sender meant			
What the receiver interpreted			
What information was missing? (tone, facial expression, context)			
How was it resolved?			

Why is face-to-face communication better for model synchronization than texting?

Write your response here

Part 3: Empathy as Prediction

Goal: Practice using your generative model of another person's internal states.

Watch a short video clip (a movie scene, a news interview, or a conversation between friends) with the sound off. Based only on visual cues:

1. What emotions do you think each person is feeling? Why?
2. What internal states are you inferring (latent states you cannot observe directly)?
3. Now watch with sound. How accurate were your predictions?
4. What does this tell you about empathy as a form of social inference?

Write your response here

Part 4: Shared Narrative Exercise

Goal: Build a shared generative model with a partner.

With a partner, take turns adding one sentence to a collaborative story. Each person must build on what the other said while staying consistent. After 10 sentences:

- 1. Did you develop a shared model of the story's world?
- 2. Where did your models of the story diverge?
- 3. How did you resolve disagreements about where the story was going?
- 4. How is this like real-world communication, where two people must align their models?

Write your response here

Summary Table

Concept	Definition	Your Example
Model Synchronization	Aligning internal models between two or more agents	
Social Inference	Predicting another person's internal states from observable behavior	
Shared Narrative	A jointly maintained generative model of a situation	
Communication Channel	The medium through which signals are transmitted (voice, text, gesture)	